Ex 2a

```
import java.util.*;
public class ex2 {
    static String encrypt(String text, int columns) {
        text = text.replace(" ", "");
        String cipher = "";
        for (int col = 0; col < columns; col++) {
            for (int i = col; i < text.length(); i = i + columns) {
                cipher = cipher + text.charAt(i);
            }
        }
        return cipher;
    }
    public static void main(String[] args) {
        String text = "A SIMPLE TRANSPOSITION";
        String result = encrypt(text, 5);
        System.out.println("Cipher: " + result);
    }
}
```

op:
Cipher: ALNISESTITPIMROOPASN

ex 2b

```
import java.util.Scanner;
public class main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the message: ");
        String message = sc.nextLine();
        message = message.replaceAll("\\s+", "").toUpperCase();
        System.out.print("Enter number of rails: ");
        int rails = sc.nextInt();
        if (rails <= 1 || rails >= message.length()) {
            System.out.println("Number of rails must be greater than 1 and less than message length.");
            return;
        }
        StringBuilder[] fence = new StringBuilder[rails];
        for (int i = 0; i < rails; i++) {
            fence[i] = new StringBuilder();
        }
        int row = 0;
        int direction = 1;
        for (int i = 0; i < message.length(); i++) {
            fence[row].append(message.charAt(i));
            if (row == 0) {
                direction = 1;
            } else if (row == rails - 1) {
                direction = -1;
            }
            row = row + direction;
        }
        StringBuilder ciphertext = new StringBuilder();
        for (int i = 0; i < rails; i++) {
            ciphertext.append(fence[i]);
        }
        System.out.println("\nOriginal Message : " + message);
        System.out.println("Encrypted Message: " + ciphertext);
        System.out.println("\nRail Fence Pattern:");
        for (int i = 0; i < rails; i++) {
            System.out.println("Rail " + (i + 1) + ": " + fence[i]);
        }
        sc.close();
    }
}
```

op:

```
Enter the message: deliver goods
Enter number of rails: 3
Original Message : DELIVERGOODS
Encrypted Message: DVOEIEGOSLRD
Rail Fence Pattern:
Rail 1: DVO
Rail 2: EIEGOS
Rail 3: LRD
```